Green Bond Market in Asia -

A Country-Wise Breakdown

The green bond market in Asia has grown into a mature market for financing needs concerning sustainability in the region. These, who were made to fund projects with environmental or climate concerns, increased significantly in the past ten years as countries have pledged to meet their climatological objectives. Following is an analysis of green bond issuance patterns in countries, their amounts, leading countries and their issuers, and key sectors that were financed through these instruments.

1. **China: The Most Important Market for Green Bonds**

In terms of the Asian green bond markets, China is unquestioningly the leader taking over a staggering 60% of the total issue in the region. In 2023, China's issuance of green bonds reached $67 Their economic drive hinge on mitigating carbon emissions and steering towards a low carbon economy hence the issuance of such green bonds is growing exponentially The concentration is expected to grow at a higher rate than ever.

Key Issuers:

- SOEs. Among other things, key issuers of foreign bonds include large state banks in China, for example, Industrial and Commercial Bank of China (ICBC) and China Development Bank (CDB). These commercial banks have backed bonds worth several billions of dollars most of which have however been raised strictly to finance investments in renewable sources of energy ie solar, wind and hydro energy.
- Corporations: Apart from the utilities and industrial segments, large corporations are frequent issuers too. For instance, State Grid Corporation of China issued green bonds for renewable energy projects. In the case of China National Petroleum Corporation (CNPC) bonds were issued also in order to support their investment in more environmentally friendly technologies in the fossil fuel industry and thus, assist in the transition to further greener model.
- Government Support: Aiding these measures is the China government who provides the necessary orientations through frameworks such as the "Green Bond Endorsed Project Catalogue" which outlines the criteria for projects that can be classified as green. Such measures stimulate the development of the green bond market since it defines what a green project is and what it is not.

Sectors Funded:

- Renewable Energy: Solar, wind and hydropower are the prevalent sectors.
- Pollution Prevention: With aggressive targets concerning air and water quality, China will make good use of green bonds in raising finances for technology aimed at pollution reduction.
- Sustainable Transport: Development of electric vehicles (EV) infrastructure and adoption of high-speed trains are among the key areas.

The activity of issuing green bonds has been the most pronounced in China regarding volume as this country is leading the way in global climate action and it can be expected that even more green bonds will be issued by the country as well as others in the effort towards carbon neutrality by the year 2060.

2. **Japan: An Expanding Green Bond Frontier Market**

In Asia, the market of green bond is expanding and in Japan, the third position has been taken by the green bond issuance. As at June 2023, the primary market green bond issuances in Japan was about 20 billion dollars, which can be attributed to the business activities of its corporations and government-backed projects.

Key Issuers:

- Government Bonds: In terms of recently sovereign green bonds issued, few regions come close to Japan. Also, it has announced more green bonds such as a much awaited low carbon economy bond of 2 trillion yen (11 billion US dollars).
- Corporate Bonds: Corporations such as Toyota and MUFG are among the significant green bonds perpetrators. Green bonds issued by these corporations were, however, on average, about 10-15 billion yen; the proceeds were used to fund clean energy and sustainable infrastructure initiatives.
- Banks: Sumitomo Mitsui Trust Bank and other banking institutions in Japan have also been active causes of Green bond issuance for the purpose of sustaining their investment on renewable energy and energy efficiency technology projects

Sectors Funded:

- Renewable Energy: Investments in solar and electricity from wind farm installations at sea.
- Efficient Use of Energy: The ones which improve the energy use in commercial buildings and production plants.
- Sustainable Transport: Electric Automotive initiatives and embedded green mass transport systems are two of the green projects in Japan that have received the most financing.

The latter is driven, in part, by Japan's own goal of achieving carbon neutrality in 2050 and therefore, the Japanese green bond market is expected to continue expanding.

3. South Korea: Focus on Transition Bonds

The growth of green bonds and sustainable development in South Korea has been particularly high since 2017. Total issuances in 2023 exploded to over 10 billion USD. This country is unique among them in that it offers a narrower focus: transition bonds or those that assist high carbon intensity industries in their low carbon transitions.

Key Issuers:

- Corporations: Enterprises such as KHNP and KEPCO have floated green bonds to develop solar and wind as well as hydrogen generation capabilities.
- Government Bonds: Sovereign bonds have been issued in south Korea too specifically under the green new deal bold and affordable housing initiatives to fund green economic projects at scale.
- Banks: As part of their efforts in financing for green housing and urban development, South Korean banks like Shinhan Bank and Woori Bank have issued bonds.

Sectors Funded:

- Renewable Energy: Sectors within wind, solar and hydrogen energy provide significant opportunities that can be funded by the issuance of green bonds.
- Energy Efficiency: Sectors which include retrofitting existing structures to more energy-saving standards.
- Pollution Control: Mostly employed for issues dealing with air quality and waste in green bonds, South Korea has largely beneficiaries of these financial instruments.

The Green New Deal of the South Korean government promises to increase the issuance of green bonds, which will be used mainly for the purpose of decarbonizing heavy industries.

4. India: Growing Green Finance Market

India has witnessed an upsurge in the green finance market India's green bond market, although considerably smaller than those of China or Japan, has however faired comparatively well. In 2023, India issued green bonds worth $8 billion, and this tendency is expected to escalate in the coming years as the country aims to achieve its renewable energy and carbon reduction targets.

Key Issuers:

- Public Sector Banks: The Indian Renewable Energy Development Agency (IREDA) as well as optimally State Bank Of India (SBI) have been one of the active issuers investing their funds in solar and wind energy projects.
- Private Sector: Players like Adani Green Energy as well Tata Power too have come out with issues in the green bond market for the purpose of raising low-cost capital for their projects in clean energy and related technologies.

Sectors Funded:

- Renewable Energy: Most of the bonds raised are related to construction of wind and solar projects as India has set an ambitious target of generating 500GW of renewable energy capacity by 2030.
- Electric Vehicles: Infrastructure for electric vehicles is being developed alongside their manufacturing.
- Clean Technology: In India, the Funds that have been raised by the issuance of green bonds have found application in waste management systems, de-watering systems, clean cooking stoves etc. - only to name a few.

As India has put in place an ambitious target with regards to renewable energy, it is fairly certain that the green bonds market of India will boom at an even greater scale in order to fulfill the essential capital for accomplishing such climate goals.

5. Southeast Asia: Emerging Markets

The Green Bond market is relatively new to some Southeast Asian countries but Singapore, Indonesia, and Malaysia are stepping up towards the green finance market.

The country has also issued Islamic green bonds or green sukuks worth $3billion. These sukuks which are mainly renewable energy, sustainable agriculture, and forest conservation initiatives. There is a growing market for green sukuks in Indonesia as the government is looking at the Dual Financing of Sukuk Bonds and Environmental Architecture Integration.

Singapore has also been active in the green bond range as it has very sound laws regarding sustainable finance raising $4.6 Billion worth of green bonds. Singapore's Monetary Authority has developed some rules and regulations that have encouraged the private sector significantly. These are used mainly to finance clean energy, green structure and urban development projects.

There was also some progress in Malaysia that managed to issue green bonds worth $2 billion aimed at financing energy-saving buildings and renewable energy schemes.

Key Issuers in these countries include:

- Sovereign Issuers: Indonesia has launched several sovereign green sukuks. Also, Malaysia has issued its own sovereign green sukuk.
- Corporates: Major private placements are from energy, real estate developments, and infrastructure segments from the likes of Petronas, SP Setia and others.

1. **Green Bond Issuance by Country (2023)**

Country	Issuance Amount (in billion USD)
China	67
Japan	20
South Korea	10
India	8
Southeast Asia	Varies (approx. 3 for Indonesia's sukuks, 4.6 for Singapore, 2 for Malaysia)

2. **Key Sectors Funded in Each Country**

Country	Renewable Energy	Pollution Control	Sustainable Transport	Energy Efficiency	Clean Technology	Other
China	High (solar, wind, hydro)	High (air & water quality)	High (EV infrastructure, trains)	Moderate	Low	N/A
Japan	High (solar, wind, offshore)	Low	High (EVs, mass transit)	Moderate	Low	N/A
South Korea	High (solar, wind, hydrogen)	Moderate	Moderate (EVs)	High	Low	N/A
India	High (solar, wind)	Low	Moderate (EV infrastructure)	Low	High (waste management, clean cooking)	N/A
Southeast Asia	High (renewable energy)	Low	Moderate	Moderate	High (sustainable agriculture, forest conservation)	N/A

3. **Issuers by Category (Approximate Distribution)**

Country	SOEs	Corporations	Banks	Government
China	High	Moderate	High	High
Japan	Low	High	Moderate	High
South Korea	Low	High	Moderate	High
India	Moderate	High	Moderate	Low

Country	SOEs	Corporations	Banks	Government
Southeast Asia	Moderate	High	Low	Moderate

4. Green Sukuks vs. Conventional Green Bonds in Southeast Asia

Country	Green Sukuks (in billion USD)	Conventional Green Bonds (in billion USD)
Indonesia	3	N/A
Malaysia	N/A	2
Singapore	N/A	4.6

5. Timeline of Green Bond Growth (Asia)

Year	China	Japan	South Korea	India	Southeast Asia
2013	1.5	0.3	0.1	0.2	N/A
2014	3.0	0.5	0.2	0.3	N/A
2015	6.0	1.0	0.3	0.4	N/A
2020	50.0	15.0	5.0	6.0	2.0
2023	67.0	20.0	10.0	8.0	9.6

6. Top Green Bond Projects by Sector

Sector	China	Japan	South Korea	India	Southeast Asia
Renewable Energy	High	High	High	High	High
Pollution Control	High	Low	Moderate	Low	Low
Sustainable Transport	High	High	Moderate	Moderate	Moderate
Energy Efficiency	Moderate	Moderate	High	Low	Moderate
Clean Technology	Low	Low	Low	High	High

7. **Carbon Neutrality Targets vs. Issuance Levels**

Country	Carbon Neutrality Target Year	Issuance Amount (in billion USD)
China	2060	67
Japan	2050	20
South Korea	2050	10
India	2070	8
Southeast Asia	Varies by Country	Varies (approx. 9.6 billion USD)

Reference:

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